

第四章作业

4.15

① 仅考虑 20V 电压源:

$$U_1 = 20 \times \frac{\frac{1}{\frac{1}{4} + \frac{1}{4}}}{2 + \frac{1}{\frac{1}{4} + \frac{1}{4}}} = 10V$$

$$i_1 = \frac{U_1}{4} = 2.5A$$

② 仅考虑 16V 电压源:

$$U_2 = -16 \times \frac{\frac{1}{\frac{1}{4} + \frac{1}{4}}}{4 + \frac{1}{\frac{1}{4} + \frac{1}{4}}} = -4V$$

$$i_2 = \frac{U_2}{4} = -1A$$

③ 仅考虑 2A 电流源:

$$2 \parallel 4 = \frac{4}{3}, \quad 3 + \frac{4}{3} = \frac{13}{3}$$

$$i_3 = \frac{1}{1 + \frac{13}{3}} \times 2 = \frac{3}{8} = 0.375A$$

则叠加后 $i = i_1 + i_2 + i_3 = 2.5 + 0.375 - 1 = 1.875A$

$$P = i^2 R = (1.875)^2 \times 3 = 10.55W$$

4.31

将受控电压源转为受控电流源, 再转为受控电压源, 则

$$V_x = 3i,$$

使用 KVL:

$$-12 + (3 + \frac{24}{5})i + \frac{24}{21}V_x = 0.$$

可解得:

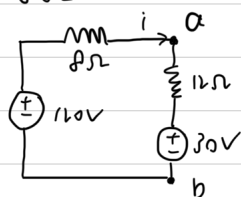
$$V_x = 3.652V$$

4.36

可知 12Ω 电阻两端的等效电压为 $V_{Th} = 150 \times \frac{40}{10+40} = 120V$

而 $R_{Th} = 40 \parallel 10 = 8\Omega$

因此, 其变成:



使用 KVL, 则

$$120 - 8i - 12i - 30 = 0$$

$$\text{可得 } i = 3A$$

4.57

将电压源短路, 则在受控电流源上方节点 A, 有

$$i + 0.5V_x = \frac{1}{10} + \frac{1-V_x}{2}, \text{ 得 } i + V_x = 0.6$$

在 6Ω 上方节点 B, 有

$$\frac{1-V_x}{2} = \frac{V_x}{3} + \frac{V_x}{6}, \text{ 得 } V_x = 0.5$$

则可得 $i = 0.1, R_{Th} = \frac{1}{i} = 10\Omega$

求 V_{Th} , 设 6Ω 上方电压为 V_1 , 受控电流源上方电压为 V_2 ,

$$\text{则有 } \begin{cases} \frac{50 - V_1}{2} = \frac{V_1}{6} + \frac{V_1 - V_2}{1} \\ 0.5V_x + \frac{V_1 - V_2}{2} = \frac{V_2}{10} \end{cases}$$

则可解得 $V_{Th} = 166.67 \text{ V}$,

$$I_N = \frac{V_{Th}}{R_{Th}} = 16.667 \text{ A}$$

$$R_N = R_{Th} = 10 \Omega$$

4.67

将除 R 以外的电路求等效电路,

$$R_{Th} = 20 \parallel 20 + 10 \parallel 90 = 25 \Omega$$

设上方节点, 下方节点, 左边中点, 右边中点电压为 V_T, V_B, V_L, V_R .

$$\begin{cases} \frac{V_T - V_L}{20} + \frac{V_T - V_R}{20} = 0 \\ \frac{V_L - V_T}{20} + \frac{V_L - 0}{10} + \frac{V_R - V_T}{20} + \frac{V_R - 0}{90} = 0 \\ V_L - V_R = 100 \end{cases} \quad \text{得 } |V_{Th}| = |V_T - V_B| = 70 \text{ V}$$

(a) 吸收最大功率时 R 为 25Ω

$$(b) P_{max} = \frac{V_{Th}^2}{4R_{Th}} = 49$$